Software Requirements Specification (SRS)

Project Title

Secure Medical Data Transmission Using Hybrid Cryptography and DWT Steganography

1. Introduction

1.1 Purpose

The purpose of this project is to develop a secure web-based system for transmitting patient medical information between authorized hospitals and doctors. The system protects sensitive medical data using Hybrid Cryptography (AES + RSA), SHA-256 hashing for integrity verification, and Discrete Wavelet Transform (DWT) Steganography for secure data hiding inside medical images.

1.2 Scope

The system enables hospitals to securely exchange patient medical reports over the Internet.

The project provides:

- Secure doctor authentication
- Hospital management
- Medical report encryption
- SHA-256 integrity verification
- DWT image steganography
- Secure report transmission
- Secure extraction and decryption of received reports

Only authorized administrators and doctors can access the system.

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2. Overall Description

2.1 Product Perspective

The system is a web-based application accessible through a browser.

Two user roles are available:

- Administrator
- Doctor

The Administrator manages hospitals and doctor accounts.

Doctors securely send and receive patient medical reports.

2.2 Product Functions

Administrator

- Login securely
- Add hospitals
- Manage hospitals
- Add doctors
- Manage doctor accounts
- View uploaded reports
- Manage system users
- Logout

Doctor

- Login securely
- Upload patient medical reports
- Encrypt reports using AES
- Encrypt AES key using RSA
- Generate SHA-256 hash
- Embed encrypted report into medical image using DWT
- Send secure report
- Receive secure report
- Extract hidden data
- Verify integrity
- Decrypt report
- View original medical report
- Logout

2.3 User Classes

Administrator

Responsible for system management.

Doctor

Responsible for secure transmission of patient reports.

2.4 Operating Environment

Operating System:

- Windows 10/11

Frontend:

- HTML
- CSS
- JavaScript

Backend:

- Python (Flask)

Database:

- MySQL

Browser:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

2.5 Constraints

- Internet connection required
- Only registered users can log in
- RSA keys must remain confidential
- Medical images should support DWT embedding
- Reports cannot be accessed without authorization

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3. Functional Requirements

FR1 Home Page

- Display project information
- Display Login button

FR2 Login Module

- Admin login
- Doctor login
- Username validation
- Password validation

FR3 Administrator Module

- Add hospital
- Edit hospital
- Delete hospital
- Add doctor
- Edit doctor
- Delete doctor
- View reports

FR4 Doctor Module

- Upload patient report
- Enter patient details
- Select receiver hospital
- Select receiver doctor

FR5 Encryption Module

- Encrypt report using AES
- Encrypt AES key using RSA
- Generate SHA-256 hash

FR6 DWT Steganography Module

- Upload cover image
- Embed encrypted report
- Generate stego image
- Save stego image

FR7 Transmission Module

- Send stego image securely
- Store transmission details

FR8 Receiver Module

- Receive stego image
- Extract hidden data
- Verify SHA-256 hash
- Decrypt report
- Display original report

FR9 Logout Module

- Securely end user session

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4. Non-Functional Requirements

Security

- AES encryption
- RSA encryption
- SHA-256 hashing
- Secure authentication
- Authorized access only

Performance

- Fast encryption and decryption
- Efficient image processing
- Low transmission delay

Reliability

- Accurate data transmission
- Secure storage
- Consistent system performance

Usability

- User-friendly interface
- Easy navigation
- Responsive design

Availability

- Accessible anytime through the web

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5. System Architecture

Home Page

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Login Page

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Admin / Doctor Login

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Dashboard

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Upload Report

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AES Encryption

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RSA Encryption

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SHA-256 Hash

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DWT Steganography

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Send Stego Image

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Receiving Doctor

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Extract Hidden Data

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Verify Hash

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Decrypt Report

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View Original Medical Report

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6. Hardware Requirements

- Intel Core i3 Processor or higher
- 4 GB RAM (Minimum)
- 8 GB RAM (Recommended)
- 20 GB Free Disk Space

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7. Software Requirements

- Windows 10/11
- Python 3.x
- Flask
- MySQL
- HTML
- CSS
- JavaScript

- OpenCV
- NumPy
- PyCryptodome
- Pillow

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8. Future Enhancements

- Patient Portal
- Mobile Application
- Cloud Storage
- Multi-Factor Authentication
- Audit Logs
- AI-based Threat Detection
- Digital Signature Support

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9. Conclusion

The Secure Medical Data Transmission System provides a reliable and secure platform for exchanging patient medical information between hospitals. By combining AES, RSA, SHA-256, and DWT Steganography, the system ensures confidentiality, integrity, and secure transmission of sensitive medical data while allowing only authorized users to access patient information.